Model	Cost Matrix ($C_{i,j} = 1$)	Overall Certified Robust Accuracy	Cost-Sensitive Robust Accuracy
Overall Robust	for all i, j	72.7%	72.7%
Even-seeds Robust	$i \in \{0, 2, 4, 6, 8\}$	38.0%	77.5%
Odd-targets Robust	$j \in \{1, 3, 5, 7, 9\}$	21.1%	86.5%
Seeds (2,3,5,6,8) Robust	$i \in \{2, 3, 5, 6, 8\}$	38.1%	74.0%
Targets (0,1,4,7,9) Robust	$j \in \{0, 1, 4, 7, 9\}$	11.1%	89.7%
Seed-modulo-5 = 0 Robust	$i \in \{0, 5\}$	16.7%	88.0%
Target-modulo-5 = 3 Robust	$j \in \{3, 8\}$	8.3%	94.0%
Seeds (3,5) Robust	$i \in \{3, 5\}$	15.9%	81.1%
Targets (1,7) Robust	$j \in \{1, 7\}$	1.4%	97.0%
Seed-modulo-10 = 3 Robust	$i \in \{3\}$	8.5%	84.2%
Target-modulo-10 = 7 Robust	$j \in \{7\}$	0.2%	98.4%

Table 1: Models trained using cost-sensitive robustness for use in ensembles. One representative model is shown from each ensemble for the sake of brevity. We show the robust cost-matrix for each model by listing the i and j values where $C_{i,j} = 1$ ($C_{i,j} = 0$ for all others), as well as its overall robust and cost-sensitive accuracy.

4. Adversarially-clustered target digits robust.

- Ten model ensembles:
  1. Seed digits robust models.
  2. Target digits robust models.

A representative selection of different models we use are described in Table 5.1. The overall robust model is a single model trained to be robust on all seed-target pairs (this is the same as standard certifiable robustness training using the convex adversarial polytope). The other models were trained using different cost-matrices. These cost-matrices are shown in Table 5.1. All these models had the same architecture, and they were trained on L_∞ distance of 0.1. Each model had 3 linear and 2 convolutional layers.

The adversarial clustering was done to ensure digits that appear visually most similar to each other are grouped together. This similarity between a pair of digits was measured in terms of how easily either digit of the pair can be adversarially targeted to the other digit. These results are consistent with our intuitions about visual similarity — for example, MNIST digits 2, 3, 5, 8 are visually quite similar, and we also found them to be adversarially similar, hence clustered together.

5.2 Certifying using MIPVerify

We used the MIP verifier on two shallow MNIST networks. One of the networks had two fully-connected layers, and the other had three fully-connected layers. The two-layer network was trained to be robust on even-seeds and the three-layer network on odd-seeds. We used adversarial training using PGD attacks to robustly train the models. Even with adversarial training, however, the models were not really robust. Even at L_∞ perturbation of $\epsilon = 0.02$, which is very low for MNIST dataset, the models only had robust accuracy of 23% and 28% respectively. The reason for the lack of robustness is because the networks were very shallow and lacked a convolutional layer. We could not make the models more complex because doing so makes the robust certification too performance-intensive. Still, even with these non-robust models, we can see some interesting results for the ensemble of the two models. We discuss them below.

To understand the robustness possible by constructing an ensemble of the two models, we compute the minimal L_1 adversarial perturbation for 100 test seeds for the two single networks and the ensemble average network built from them. We used L_1 distance because the MIP verifier performs better with this, due its linear nature, compared to L_2 or L_∞ distances. More than 90% of the seeds were verified within 240 seconds. Figure 2 shows the number of seeds that can be proven robust using MIP verification at a given L_1 distance for each model independently, the maximum of the two models, and the ensemble average model. The verifier was not always able to find the minimal necessary

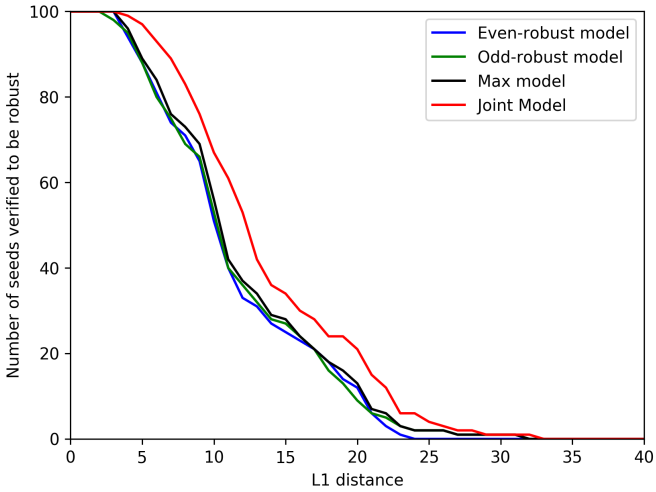


Figure 2: Number of test seeds certified to be robust by the single models and the ensemble average model for different L_1 perturbation distance constraints.

perturbation for the ensemble network within the time limit of 240 seconds. In those cases, we reported the maximum adversarial distance proven to be safe at the time when time limit exceeded – which respresents an upper bound of minimal adversarial perturbation. We note from the figure that number of examples certified by the ensemble average model is higher than that for either individual model at all minimal L_1 distances.

In general, though, we found the MIP Verificaton does not scale well with networks complex enough to be useful in practice. Deeper networks and use of convolutional layers makes the performance of MIP Verify significantly worse. Furthermore, we found that robust networks were harder to verify than non-robust networks with this framework. Because of this, we decided not to use this approach for the remaining experiments which is more practical networks.

5.3 Convex adversarial polytope certification

As the MIP verification does not scale well to larger networks, for our remaining experiments we use the convex adversarial polytope formulation by Wong et al. [29]. We conduct experiments with ensembles of two, five, and ten models, using the models described in Table 5.1. Table 2 summarizes the results.

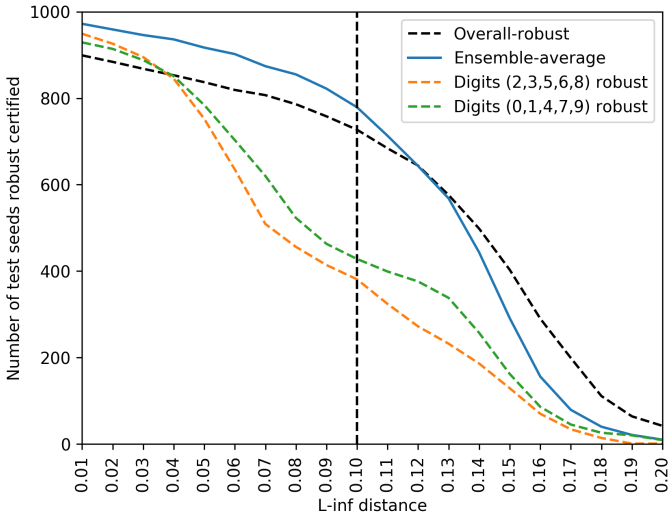
Joint robustness of two-model ensembles. We evaluated two-model ensembles with different choices for the models, using the three composition methods. We ensured that the averaging ensemble could be treated as a single sequential model made of fully-connected linear layers, so that the robust verification formulation was still valid when applied on it. To do this, we had to first convert the convolutional layers of the single models into linear layers, and then the linear layers of the two models were combined to create larger linear layers for the joint model. We can then calculate the robust error rates of the ensemble average model, as well as the unanimity and majority ensembles for the two-model ensemble. The key here is that no changes were needed to be made to the existing verification framework.

Table 2 shows each ensemble's robust accuracy. For two-model ensembles, the unanimity and the majority frameworks are the same. Thus we can use the same ensemble average technique to certify them. For adversarial clustering into 2-models, we used two clusters – one for digits (2, 3, 5, 6, 8) and the other for digits (0, 1, 4, 7, 9).

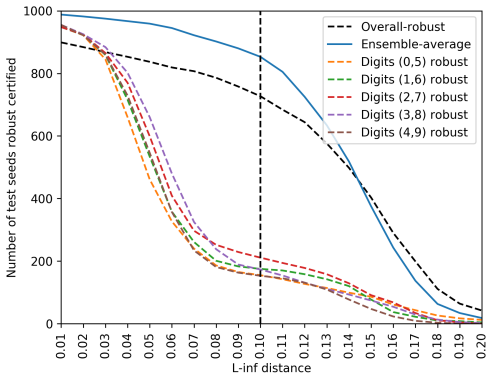
Compared to the single overall robust model, where 72.7% of the test examples can be certified robust, with two-model ensembles we can certify up to 78.1% of seeds as robust (using the averaging composition with the adversarially clustered seed robust models).

Models	Composition	Certified Robust	Normal Test Error	Rejection
Overall Robust	Single	72.7%	5.0%	-
Even/Odd-seed	Unanimity	74.7%	1.3%	5.0%
	Average	75.9%	3.3%	-
Clustered seed (2)	Unanimity	77.3%	1.5%	6.0%
	Average	78.1%	3.0%	-
Seed-modulo-5	Unanimity	84.1%	0.3%	8.1%
	Average	85.3%	1.7%	-
Clustered seed (5)	Unanimity	83.8%	0.7%	7.1%
	Average	84.3%	1.4%	-
Seed-modulo-10	Unanimity	85.4%	0.1%	9.7%
	Average	85.6%	1.5%	-

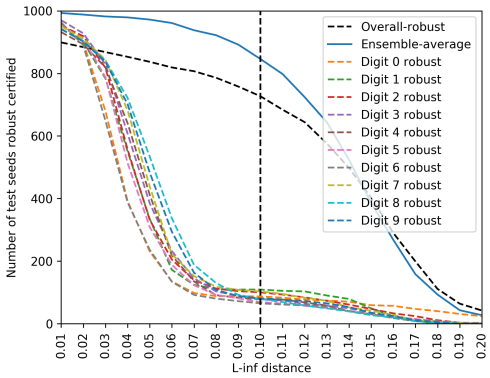
Table 2: Robust certification and normal test error and rejection rates for single model and two, five, and ten-model ensembles for L_∞ adversarial examples with $\epsilon = 0.1$.



(a) Cluster Two-Model Ensemble



(b) Modulo-5 Seeds Ensemble



(c) Ten-model ensemble

Figure 3: Number of test examples certified to be jointly robust using each model ensemble for different ϵ values.